College of Business
香港城市大學
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FinTech and AI in Finance

Class 4: Nonlinear Machine Learning for Return Prediction

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Semester A, 2026/27

City University of Hong Kong

Nonlinear ML for the Same Financial Questions

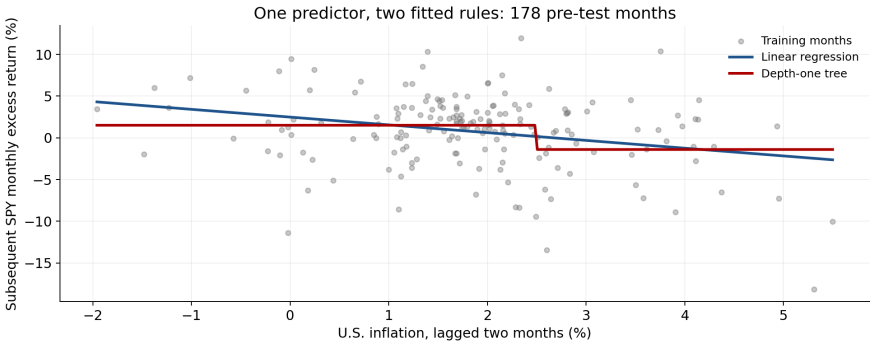
Class 3 used linear combinations of predictors. Here we allow a signal's effect to change with its level or with other characteristics.

- **Trees:** forecasts differ across observable regimes.
- **Tree ensembles:** forests and boosted trees combine trees to stabilize or refine a forecast.
- **Neural networks:** learn combinations of signals and their interactions.

We compare market excess-return forecasts and weekly stock excess-return forecasts using Class 3's dates and benchmarks.

Regimes and Nonlinear Forecasts

Linear and Tree Forecasts from the Same Data



Keep in mind

A line uses one slope; a tree uses discrete regimes. The fitted rule must predict well on later data.

Regression Trees: Forecasts by Regime

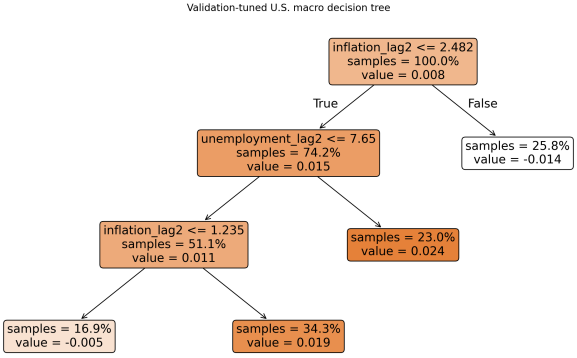
A tree divides observations into groups called **leaves**. In leaf R_ℓ, the forecast is the mean training return:

$$\widehat{c}_\ell = \frac{1}{n_\ell} \sum_{i \in R_\ell} y_i.$$

n_ℓ: number of training observations in the leaf.

- **When useful:** the relation changes at a threshold or depends on another predictor.
- **How to forecast:** follow the fitted splits into one leaf and use its mean return.
- **Complexity:** depth and minimum leaf size limit how finely the tree divides the data.

A Single Regression Tree: How the Splits Work



At each node, $\leq$ goes left and $>$ goes right. Continue until a terminal leaf, whose mean training excess return is the forecast.

Regression Trees: Fitting and Using the Rule

Split criterion: try predictors j and thresholds s. Predict each group's mean and choose the smallest total squared error:

$$(j^*, s^*) = \arg\min_{j,s} \left[\sum_{i \in R_L(j,s)} (y_i - \bar{y}_L)^2 + \sum_{i \in R_R(j,s)} (y_i - \bar{y}_R)^2 \right].$$

R_L: node observations with $x_{ij} \leq s$; R_R: those with $x_{ij} > s$. Each $\bar{y}$ is the corresponding training mean.

In the preceding tree, the first selected rule is lagged inflation $\leq 2.48\%$.

1. Repeat within each child node, subject to depth and leaf-size limits.
2. Choose those limits on chronological validation data; refit before testing.
3. Route a new observation using its predictors only.

How to Read the First Split

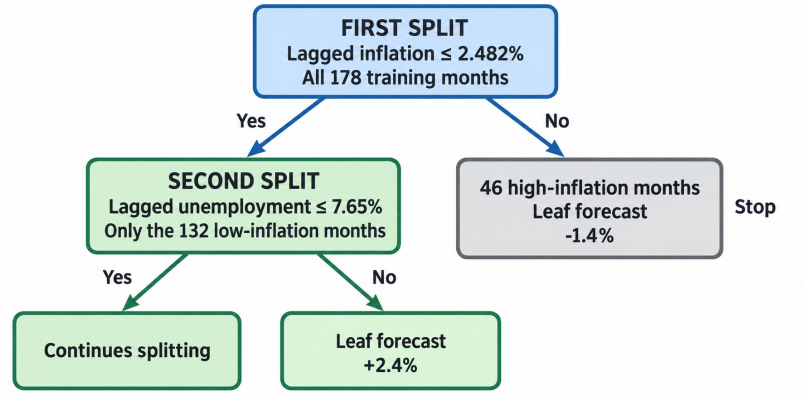
The validation-tuned U.S. tree asks whether two-month-lagged inflation is at most 2.48%.

Training region	Observations	Mean excess return
Inflation $\leq 2.48\%$	132 months	$+1.5\%$
Inflation $> 2.48\%$	46 months	-1.4%

These are first-level node means. The left node splits further, so its $+1.5\%$ is not the final leaf forecast.

The fitted threshold describes a predictive relation, not a causal effect of inflation.

The First and Second Splits



The second split applies the same squared-error criterion **within a child node**.

U.S. market model shown on page 6. Predictors lagged two months. Leaf forecasts are next-month excess returns.

Cross-Sectional Splits and Time-Series Splits

Cross-section: split stocks

In the same month, separate firms by a characteristic such as size.

Different stock groups can have different return relations.

Time series: split months

For the same market, separate months by a state such as lagged inflation.

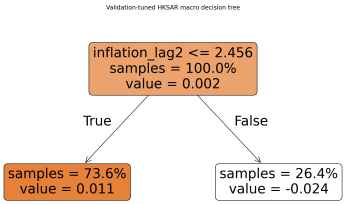
Different market states can have different return relations.

Asset Heterogeneity and Uncommon Factors

The paper combines stock-group trees with variable selection: a characteristic can enter the return model for some groups but not others. Separate expansion and recession fits compare these patterns across business-cycle states.

Cong, Feng, He, Li, and Zhang (2026 working paper). The business-cycle comparison uses retrospective NBER dates.

Similar Inflation Thresholds, Different Market Outcomes



Validation again picks depth 1 with 36-month leaves, and the root again splits on U.S. inflation at 2.46%. The 131 training months at or below it forecast +1.1% for the Hang Seng; the 47 months above it forecast -2.4%.

Keep in mind

The U.S. tree has test OOS $R^2 = -2.6\%$ versus MA(12), compared with +4.7% for HKSAR. Similar fitted thresholds can have different forecasting value.

Regression Trees: Choosing Depth on Validation Data

Suppose we fit trees with different depth limits on the same development months:

Maximum depth	1	2	4	8
Training MSE $\times 10^4$	24	21	16	9
Validation MSE $\times 10^4$	28	**26**	29	35

Choose depth two. Deeper trees fit the development returns more closely but make worse forecasts on the validation months.

Refit at the chosen depth before the test. Minimum leaf size provides a second way to limit how finely the tree divides the sample.

Illustrative errors, not results from the course dataset.

Tree Depth and Financial Regimes

- **Depth one:** one question gives two return forecasts.
- **Depth two:** a second question can distinguish momentum behavior within a volatility regime.
- **Greater depth:** narrower groups describe more local patterns, using fewer observations per group.

Validation chooses how many distinctions the data can support.

Leaf Size Sets the Evidence Required for a Regime

- **Small leaf:** can represent a rare crisis or liquidity state, but its average return is noisy.
- **Large leaf:** makes each regime estimate more stable, but may hide a real rare state.
- **In our market exercise:** candidate minimum leaf sizes are 12, 24, and 36 months. Validation chooses among them.

Keep in mind

A high-return leaf based on a few months is a noisier estimate than a similar leaf observed repeatedly.

Worked Example: Predicting with Two Leaves

Six training months are split at lagged inflation of 2.5%:

Inflation	1.0%	1.8%	2.2%	3.0%	3.6%	4.4%
Next-month return	3%	1%	2%	0%	−1%	−2%

$$\widehat{r}_{\text{low}} = (3 + 1 + 2)/3 = 2\%, \qquad \widehat{r}_{\text{high}} = (0 - 1 - 2)/3 = -1\%.$$

A new month with lagged inflation of 2.0% receives a 2% forecast. At 3.5% inflation, the forecast is −1%.

The lagged inflation observation determines the branch. The leaf average comes from training returns.

Illustrative training observations.

Worked Example: Momentum Depends on Volatility

Suppose a fitted model gives these weekly excess-return forecasts:

Market state	Low momentum	High momentum	High minus low
Low volatility	0.0%	0.6%	+0.6%
High volatility	0.0%	−0.2%	−0.2%

The momentum effect changes sign with volatility. A regression with only additive momentum and volatility terms cannot reproduce this table.

A tree can first split on volatility, then on momentum within each branch.

Illustrative fitted forecasts, not estimates from the course panel.

Using an Interaction in Stock Selection

In the preceding example, a high-momentum stock receives:

$$\widehat{r}^{e} = 0.6\% \text{ in low volatility}, \qquad \widehat{r}^{e} = -0.2\% \text{ in high volatility}.$$

A long-only manager might overweight the first stock and underweight the second.

- The same momentum screen now produces different positions across states.
- The interaction is useful if it improves later rankings or portfolio returns.

Class 5 considers how these forecasts become constrained weights.

Bagging and Boosting

Bagging: Averaging Bootstrap Fits

Bootstrap: draw n rows **with replacement** from n training rows. Some rows repeat and others are absent.

Fit the same method, the **base learner**, to each of B bootstrap samples. Average its forecasts:

$$\widehat{f}_{\text{bag}}(x) = \frac{1}{B}\sum_{b=1}^{B} \widehat{f}_b(x).$$

Each $\widehat{f}_b$ can be a **linear regression or a tree**. For predictions with variance σ_f^2 and common pairwise correlation ρ,

$$\text{Var}\left(\frac{1}{B}\sum_{b} \widehat{f}_b(x)\right) = \sigma_f^2\left[\rho + \frac{1-\rho}{B}\right].$$

Why average: reduce the uncorrelated part of prediction variance. Shared errors remain.

Use training data only. This class mainly illustrates bagging with trees.

Bagging: From Resampled Data to One Forecast

Fit the same base method to each bootstrap sample. Predict for the **same new observation**.

Suppose three fitted models forecast next month's return as

$$0.8\%, \qquad -0.2\%, \qquad 0.3\%.$$

Their bagged forecast is

$$\widehat{r} = \frac{0.8 - 0.2 + 0.3}{3}\% = 0.3\%.$$

Regression example: repeat variable selection and linear regression in each sample, then average the forecasts.

When OLS fits are stable across samples, bagging may add little. Choose model settings on later validation dates.

Illustrative forecasts.

Random Forests: Adding Random Predictor Selection

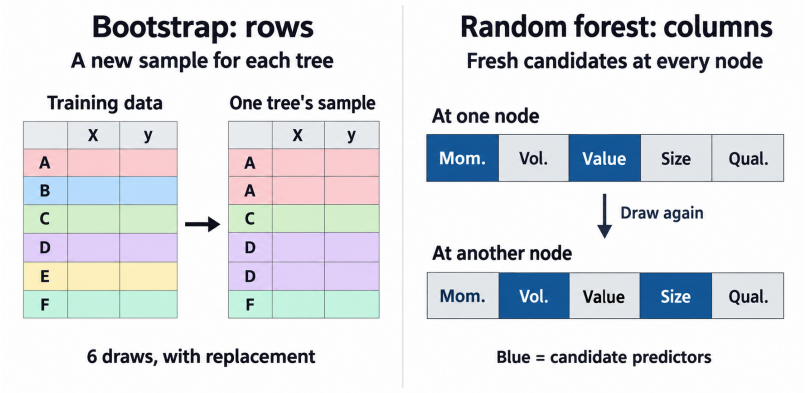
A random forest applies bagging to **trees** and adds random predictor selection at each split.

- Fit trees to different bootstrap samples of the training rows.
- At each node, search for the best split among a fresh random subset of predictors. Our model considers 5 of the 30 predictors.
- Average the B fitted trees' forecasts for a new observation:

$$\widehat{f}_{\mathrm{RF}}(x) = \frac{1}{B}\sum_{b=1}^{B}\widehat{f}_b(x).$$

Random predictor selection can reduce correlation between trees, making averaging more effective.

Bootstrap Rows and Random Predictor Subsets



Within the candidate columns, choose the **best variable and cutoff**. Use training data only.

Mom.: momentum; Vol.: volatility; Qual.: quality. Illustration: 2 of 5 variables. Course model: 5 of 30 at each split.

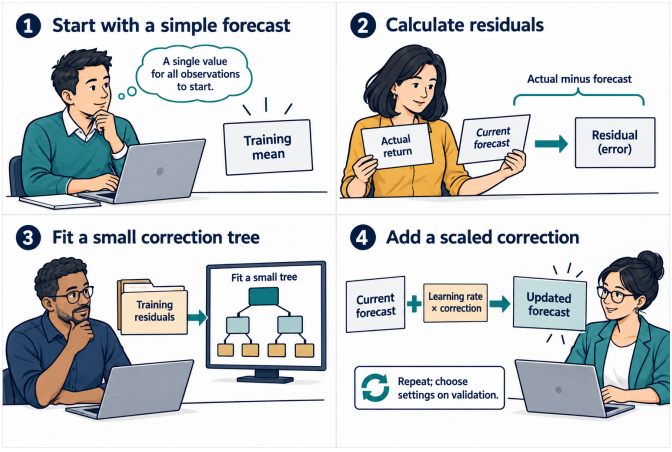
Random Forests: Fitting a Weekly Stock Model

Start with the thirty lagged stock characteristics and next-week excess returns.

1. Draw bootstrap samples from the development rows. At each split, consider a random subset of characteristics.
2. Compare tree depth and minimum leaf size on the next 26 validation weeks, keeping all stocks from each week together.
3. Refit the chosen settings on the 104 pre-test weeks and average the trees' forecasts for each test stock.

Our implementation considers $\lfloor \sqrt{30} \rfloor = 5$ characteristics per split. It uses 300 trees during validation and 500 in the final refit.

Gradient Boosting: Successive Forecast Corrections



Squared-error boosting.

$$F_0 = \bar{y}_{\text{train}},$$
$$e_{im} = y_i - F_{m-1}(x_i),$$
$$F_m = F_{m-1} + \nu h_m.$$

h_m: a base learner fitted to current training residuals.

Here we use small trees. **Simple linear regressions** are another choice.

ν: learning rate. Choose it, the number of updates, and learner complexity on validation data.

Gradient Boosting: Calculating One Update

Start from the sample mean of the same six months, $F_0 = 0.5$, and take the residuals:

Return	$+3$	$+1$	$+2$	0	-1	-2
Residual $r_t - F_0$	$+2.5$	$+0.5$	$+1.5$	-0.5	-1.5	-2.5

Fit one stump to those residuals. The 2.5% threshold wins again, with leaf means $+1.5$ and -1.5. With a learning rate $\nu = 0.1$:

$$F_1 = 0.5 + 0.1 \times (\pm 1.5) = 0.65 \text{ or } 0.35.$$

Later trees fit the remaining errors. Validation selects the number of updates.

Illustrative returns in percentage points, using the earlier six-month example.

Tune Learning Rate and Tree Count Together on Validation

- Pre-specify a small grid of learning rates and tree counts.
- Compare each pair on the same chronological validation sample.
- Prefer a stable region of good performance over one isolated minimum.
- Freeze the pair before opening the test period.

Early stopping is the same decision made one tree at a time: track validation loss as trees are added and stop when it no longer falls.

Bagging and Boosting with Different Base Learners

Two ensemble strategies. Both can use linear regressions or trees as base learners.

	Bagging	Boosting
Core idea	Refit on bootstrap samples and average	Fit successive corrections and add them
Regression example	Repeat variable selection and linear regression in each sample	Fit residuals to each predictor, choose the best, and add a small update
Tree example	Bagged trees. Random forests also sample candidate predictors	Boosted shallow trees
Main controls	Base-model complexity and ensemble size	Learning rate, update count, and learner complexity

An average or sum of linear fits **remains linear** in the same predictors. Validate any improvement on later dates.

Breiman, Bagging Predictors; Bühlmann (2006), Boosting for high-dimensional linear models.

Short-Answer Question: Is This Threshold Investable?

A tree splits at inflation $= 2.5\%$. Pre-test diagnostics:

Diagnostic	Evidence
Leaf support	Only four months support the high-inflation leaf.
Leaf-size sensitivity	Minimum leaf size 12 removes the split.
Time stability	A later pre-test refit gives a different split.

1. Identify two warning signs in the diagnostics.
2. Explain whether the threshold is credible enough to guide a financial decision.
3. Propose one additional validation check without looking at the frozen test.

Interpretability alone does not establish a stable regime.

Worked Example: Reading a Regression Tree

Suppose a fitted tree uses the following weekly forecasting rule:

Condition	Forecast return
Volatility $> 2\%$	-0.8%
Volatility $\leq 2\%$, momentum > 0	$+0.6\%$
Volatility $\leq 2\%$, momentum ≤ 0	-0.2%

A stock has volatility 1.5% and momentum -0.4%. It first enters the low-volatility branch, then the non-positive-momentum leaf:

$$1.5\% \leq 2\%, \quad -0.4\% \leq 0 \quad \Longrightarrow \quad \widehat{r} = -0.2\%.$$

The predictor value -0.4% is an input to the rule; it is not itself the forecast.

Why Random Predictor Subsets Help

Suppose nearly every bootstrap tree uses the same dominant predictor for its first split. Their forecast errors will remain highly correlated, so averaging delivers little diversification.

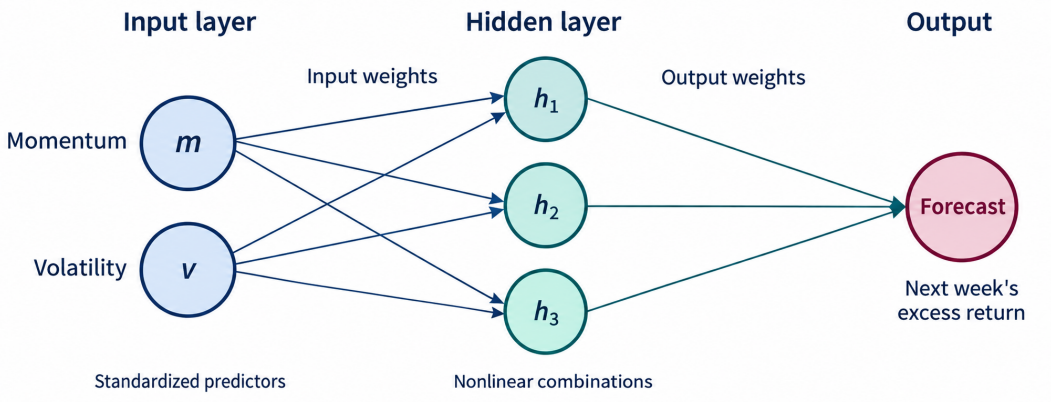
At each split, a random forest offers each tree only a subset of predictors:

- different trees can discover different useful signals;
- lower forecast correlation makes averaging more effective;
- validation is still needed to choose depth, leaf size, and feature sampling.

Randomization need not improve each tree. Its purpose is to improve the ensemble.

Neural Networks and Market Evidence

Feed-Forward Neural Networks



Inputs pass through hidden layers to the forecast, with no feedback loops. Hidden units apply nonlinear activations to weighted inputs.

Feed-Forward Networks: Why They Are Powerful

Compare a linear model on the original predictors with a network using nonlinear activations:

	Linear model	Feed-forward network
Signal response	One fixed slope per predictor	The effect can change with the predictor's level.
Interactions	Must add interaction terms explicitly	Momentum can matter differently at low and high volatility.
Combinations	Use specified predictors and transformations	Learn nonlinear combinations; deeper layers recombine earlier signals.

Nonlinear activations are essential: stacking only linear layers still gives a linear model.

Flexibility can also fit noise: use regularization and later-date validation.

Reading: Gu, Kelly, and Xiu (2020), RFS, Section 1.7.

Neural Networks: Fitting a Small Return Model

Estimate network parameters θ using training excess returns:

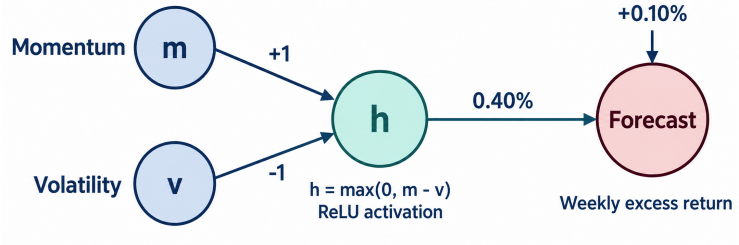
$$\widehat{\theta} = \arg\min_{\theta} \left\{ \frac{1}{2n} \sum_{i=1}^{n} \big(a_i - f_\theta(z_i)\big)^2 + \frac{\lambda}{2} \sum_{w \in \mathcal{W}} w^2 \right\}.$$

$\mathcal{W}$: connection weights; biases are unpenalized. This combines squared error with weight shrinkage.

1. Fit scaling and candidate networks on development weeks.
2. Choose hidden units, penalty, and training duration on later validation weeks.
3. Refit on pre-test weeks for the selected duration; evaluate once on the test.

Compare random initializations during development, before opening the test.

Worked Example: Reading a Hidden Unit



Stock	m	v	h	$\widehat{r}^e$
A	1.0	0.5	0.5	0.30%
B	1.0	1.5	0.0	0.10%

Higher volatility lowers this forecast, holding momentum fixed.

Illustrative weights. Inputs m and v are standardized.

Early Stopping: Reading Training and Validation Loss

Suppose a neural network produces the following losses as training continues:

Training epoch	Training MSE	Validation MSE
10	0.0025	0.0027
30	0.0021	**0.0023**
60	0.0018	0.0026

The 30-epoch fit has the lowest validation error. Additional training lowers training error but worsens prediction on later observations.

Select the stopping rule before final testing. Use chronological validation with every stock from each date kept together.

Illustrative losses, not course-panel neural-network results.

Panel Size and the Number of Independent Dates

Course example	Usable observations	Distinct dates
Market prediction	238	238 months
Stock prediction	12,281	156 weeks

Stocks in the same week share news, funding conditions, and market shocks. Thousands of rows therefore provide fewer independent market episodes than the row count suggests.

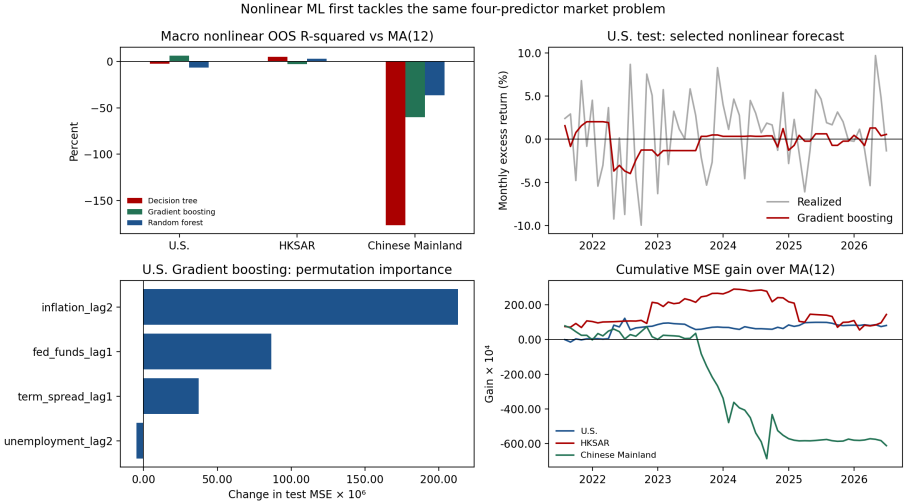
Neural networks illustrate a possible extension here. Our reported course results use linear models and tree ensembles.

Why Stock Panels Need Date-Based Splits

Thirty stocks observed over 156 weeks produce 4,680 rows, but only 156 distinct market dates.

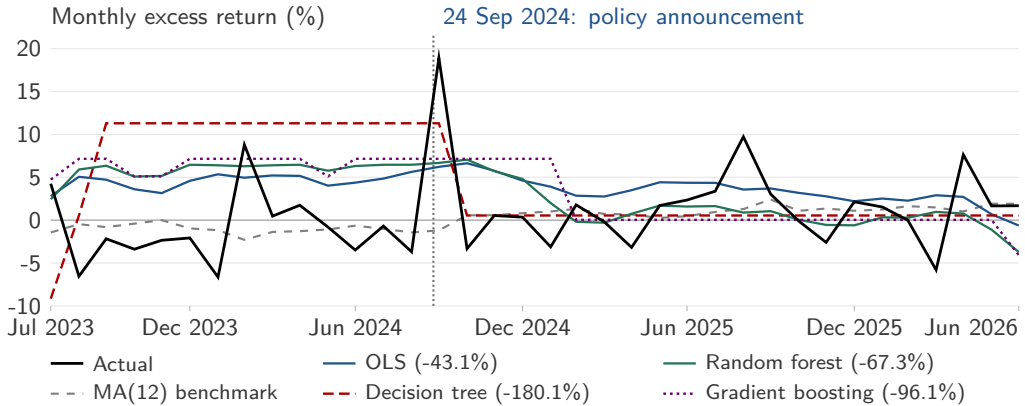
- Stocks in the same week share market news and funding conditions; the rows are not 4,680 independent market episodes.
- A random row split can train on later dates while testing on earlier dates, so it does not mimic a future forecast.
- Keep every stock from one week together. Use earlier dates for development, later dates for validation, and the final dates for testing.
- Choose network size and stopping on validation data, then compare with simpler models on the frozen test.

Nonlinear Market Excess-Return Forecasts Struggle Out of Sample



CSI 300: Policy Changes and Return Prediction

July 2023–June 2026, 36 monthly excess-return forecasts.



Models retain their June 2021 fits. Legend values: OOS R^2 relative to MA(12).

Linear and Nonlinear Market Models on a Common Test

Market	Linear selection	OOS R^2	Nonlinear selection	OOS R^2
U.S.	LASSO	-16.3%	Gradient boosting	+6.1%
HKSAR	Elastic Net	+8.4%	Decision tree	+4.7%
Chinese mainland	LASSO	-0.6%	Random forest	-36.4%

Each entry uses the family selected on that class's validation sample. Full test: July 2021–June 2026, 60 months per market, against lagged MA(12).

Added flexibility does not improve all three markets. The comparison depends on the prediction problem and the available sample.

What the Market Results Mean for an Investor

The U.S. macro predictors describe economic conditions, but their relationship with subsequent equity returns changes across markets and periods.

- A readable inflation split is easy to explain, yet may forecast poorly.
- A flat, strongly regularized forecast can outperform a detailed regime rule.
- Class 5 tests whether these forecasts change market exposure profitably.

Nonlinear Stock Return Forecasts

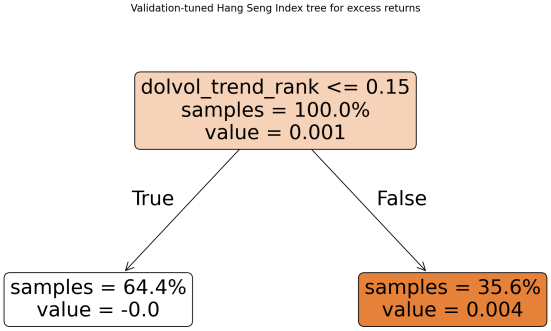
How Much Data Support a Stock-Tree Leaf?

Candidate minimum leaf sizes: 30, 60, or 120 training stock-weeks.

- A leaf groups stock-weeks with similar characteristics. Its observations can be spread across many dates.
- Several stocks from one week can share the same return shock.
- Count the distinct weeks supporting a rule as well as the number of stock-week rows.

The validation exercise keeps all stocks from the same week together.

The Stock Tree Splits on Dollar-Volume Trend



Follow the dollar-volume-trend rank to the two weekly forecasts. This is the validation-tuned tree candidate, not the overall winning nonlinear family.

Reading the Dollar-Volume-Trend Split

The fitted tree splits the centered dollar-volume-trend rank at about 0.150, roughly the 65th percentile.

Training group	Share of rows	Weekly excess-return forecast
Rank ≤ 0.150	64.4%	-0.01%
Rank > 0.150	35.6%	+0.43%

The rule associates stronger trading-activity trends with higher subsequent returns in the fitted sample. Its test OOS R^2 is -0.1% against zero.

The displayed tree is the validation-tuned tree candidate. Random forest is the selected nonlinear family.

Validation Separates Three Nonlinear Families by Very Little

Family	Depth	Min. leaf	Rate	Trees	Validation MSE $\times 10^4$
Decision tree	1	30	–	1	26.30
Random forest	2	60	–	300	26.25
Gradient boosting	1	30	0.01	150	26.64

Keep in mind

Validation selects Random forest, but all three losses are close. Report the numbers and treat small differences cautiously; the table alone provides little basis for a strong claim that one nonlinear family is superior.

Course calculation using the fixed-vintage weekly stock panel.

Multiple-Choice Question: Which Nonlinear Model Was Selected?

Three models are tuned before the frozen test period. Their mean squared errors are:

Model	Validation MSE	Test MSE
Shallow tree	0.0024	0.0019
Random forest	0.0021	0.0023
Boosting	0.0022	0.0018

Which model should be reported as the **preselected evaluation model**?

A. Shallow tree, because it has the lowest test MSE among the trees.

B. Random forest, because it has the lowest validation MSE.

C. Boosting, because it has the lowest test MSE.

D. Average the tree and boosting after observing the test results.

Discuss

Separate model selection from final evaluation. Why cannot the test-period winner replace the model frozen after validation?

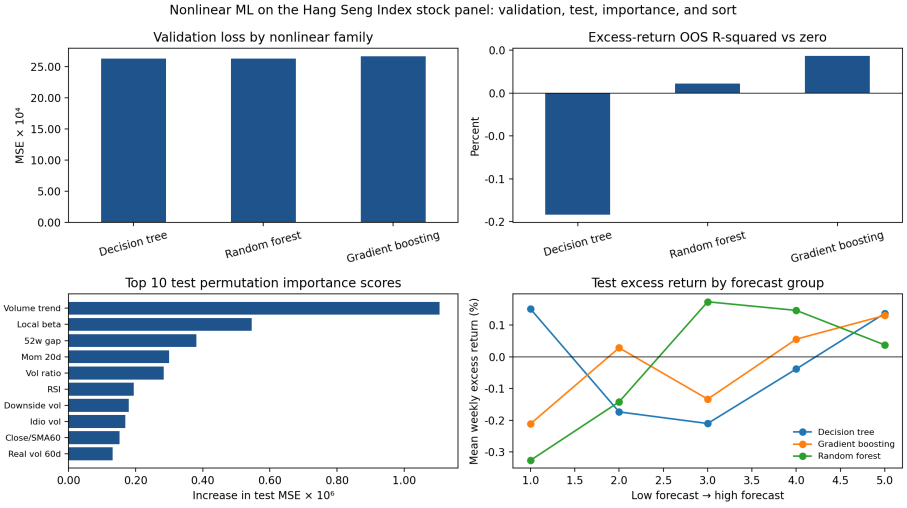
The Validation Winner Is Essentially Tied with Zero in the Test

Frozen model	Test MSE $\times 10^4$	Test OOS R^2
Random forest (selected)	21.89	+0.0%
Gradient boosting	21.88	+0.0%
Decision tree	21.92	-0.1%

Do not switch after the test

Random forest remains the validation-selected model. Its test OOS R^2 is only slightly positive, so the economic evidence remains weak.

Nonlinear Stock Return Forecasts Also Remain Weak



Worked Example: Read One Weekly Nonlinear Forecast

For Techtronic Industries (0669.HK) in the week ending 31 October 2025, returns rounded to one decimal place:

$$\underbrace{0.3\%}_{\widehat{r}^{e,\mathrm{RF}}_{i,t}} \quad \text{versus} \quad \underbrace{-8.4\% - 0.1\% = -8.5\%}_{r^{e}_{i,t}\ \text{realized}}.$$

Keep in mind

The selected model misses both sign and magnitude here. Evaluate it over all 52 test weeks, not this observation alone.

Course calculation using the fixed-vintage weekly stock panel.

Linear and Nonlinear Stock Return Forecasts Compared

Model	Role	Test MSE $\times 10^4$	OOS R^2
PCR	linear selection	21.89	+0.02%
Random forest	nonlinear selection	21.89	+0.01%
Gradient boosting	other candidate	21.88	+0.04%

Both classes use the same 52 test weeks and 4,108 stock-week outcomes. PCR is selected within the linear families, and Random forest within the nonlinear families.

In this panel, the extra flexibility does not deliver a larger test gain. It changes which characteristics drive the fitted forecast.

Stock Ranking Evidence for the Nonlinear Models

Model	Mean weekly top–bottom spread	t-statistic
Decision tree	-0.01%	-0.06
Random forest	+0.36%	1.04
Gradient boosting	+0.34%	0.87

Positive ensemble spreads have small t-statistics. Ranking and MSE evaluate different forecast properties.

52 test weeks. Equal-weight top-minus-bottom quintiles, unit exposure per side. Transaction costs ignored.

Accuracy, Ranking, and an Investment Mandate

Use of the forecast	Relevant evidence	What it assesses
Expected return	MSE, OOS R^2	numerical forecast accuracy
Stock ranking	quintile return; spread	ordering of subsequent returns
Long-only portfolio	return, risk, constraints	value after position constraints

A model can perform differently under these objectives. Choose the objective on development and validation data before the final test.

Forecast Accuracy and Stock Ranking Can Disagree

Suppose a stock model has test OOS $R^2 = -1\%$ against zero, but its top-minus-bottom portfolio earns $+0.08\%$ per week before costs.

Level accuracy

- OOS R^2 compares squared numerical errors.
- -1% means the model is slightly worse than forecasting zero for every stock.

Ranking usefulness

- The spread uses only relative ordering.
- Check statistical precision, stability, risk, concentration, and costs before investing.

Interpreting the Weak Test Gains

Test gains from additional flexibility are limited.

- Thirty technical indicators overlap substantially in the information they contain.
- A fitted threshold may describe a relationship concentrated in a short period.
- More flexibility cannot compensate for a missing predictive variable.

The next research question concerns the information set and the stability of the relationship, as well as the algorithm.